# Matt Sampson

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## About me

I am a PhD candidate at Princeton university working on developing machine learning tools for the field of computational astrophysics. I am interested in the advancement of machine learning through probability and statistics, and the advancement of science through machine learning.

## Technical Skills

| | | |
|---|---|---|
| **Languages** | : | Python (JAX, PyTorch, Flax, equinox, scipy, pandas), C++, R, MATLAB |
| **Mathematical** | : | advanced calculus and linear algebra, ODEs/PDEs, statistics and probability, Bayesian inference, computational statistics |
| **Databases/Other** | : | Git, bash, remote/cluster computing, high-performance computing, SQL |

## Experience

**Graduate Researcher (Machine Learning)** — 2022 – Present
*Princeton University* — *Princeton, NJ, USA*

- Designed, implemented, tested score-based diffusion model to assist in galaxy source separation. To be used in upcoming large telescope surveys **Python (JAX, equinox), HPC/GPU computing, remote/cluster computing**
- Implemented a method for computing efficient Hessian free $2^{nd}$ order methods to quantify hallucination from with neural networks **Python (JAX)**
- Developed automated scripts to extract data from large telescope surveys via intermediate remote clusters **bash,Python**

**Honours Researcher (Computational Astrophysics/Software)** — Jan 2021 – Nov 2021
*Australian National University* — *Canberra, ACT, Australia*

- Assisted in the development of a novel cosmic ray propagation code Criptic (**C++, Python, bash)**
- Troubleshot,fixed bugs and implemented features such as continuous time integration, external file reading, into the source code (**GDB: The GNU Project Debugger**)

**Research Assistant (Computational Statistician)** — 2019 – 2021
*Queensland University of Technology* — *Remote – Brisbane, QLD, Australia*

- Assisted in development of statistical tools improving on standard MCMC routines, increase runtime performance by **90%** with no drop model quality **MATLAB, bash**
- Designed and developed statistical analysis tools for very large datasets in biology **Python, R, MATLAB, bash**
- Worked with local cluster computers to perform large scale model fitting of multivariate models

## Education

**Princeton University** — Princeton, NJ, USA
*Ph.D. in Computational Astrophysics (machine learning)* — *2026 (Expected)*

**Princeton University** — Princeton, NJ, USA
*Masters in Computational Astrophysics (machine learning)* — *2023*

**Australian National University** — Canberra, ACT, Australia
*Honours (similar to US masters) in Computational Astrophysics* — *2021*
First class honours, GPA 6.9/7

**Queensland University of Technology** — Brisbane, QLD, Australia
*Bachelor of Mathematics (Applied and Computational)* — *2020*
Awarded with distinction GPA 7/7

**Queensland University of Technology** — Brisbane, QLD, Australia
*Bachelor of Science (Physics)* — *2020*
Awarded with distinction GPA 6.9/7

## Publications

**First author:**

1. **Sampson, M. L.**, Melchior, P., Ward, C., & Birmingham, S. (2024). “Score-matching neural networks for improved multi-band source separation" *arXiv preprint arXiv:2401.07313*
2. **Sampson, M. L.**, Beattie, J. R., Krumholz, M. R., Crocker, R. M., Federrath, C., Seta, A. (2023) “The turbulent diffusion of streaming cosmic rays through compressible, partially ionised plasma." *Monthly Notices of the Royal Astronomical Society 519 (1), 1503-1525* **Citations: 20**
3. **Sampson, M. L.**, Melchior, P. (2023) “Spotting Hallucinations in Inverse Problems with Data-Driven Priors" *ICML ML4 Astro*

**Second author / significant contributor:**

1. Krumholz, M. R., Crocker, R. M., **Sampson, M. L.,** (2022) “Cosmic Ray Interstellar Propagation Tool using Itô Calculus (CRIPTIC): software for simultaneous calculation of cosmic ray transport and observational signatures." *Monthly Notices of the Royal Astronomical Society 517 (1), 1355-1380* **Citations: 7**
2. Stevenson, S., **Sampson, M. L.**, Powell, J., et al. (2019). The impact of pair-instability mass loss on the binary black hole mass distribution. *The Astrophysical Journal, 882(2), 121*. **Citations: 128**

**Co-author**

1. Beattie, J. R., Krumholz, M. R., Federrath, C., **Sampson, M. L.,** Crocker, R. M. (2022). “Ion Alfvén velocity fluctuations and implications for the diffusion of streaming cosmic rays" *Frontiers in Astronomy and Space Sciences* **Citations: 13**

Metrics all from NASA/ADS: Total of **168 citations** - *h*-index: **4**

## Conference Presentations

1. **Sampson, M.** “Spotting hallucinations in inverse problems with data-driven priors”. *ICML workshop, Hawaii*, 2023.
2. **Sampson, M.** “Score-based diffusion models for galaxy seperation”. *Cosmic Connections, Flatiron Institute, NY*, 2023.
3. **Sampson, M.** “Turbulent diffusion of streaming cosmic rays”. *ACAMAR Meeting on Astroparticle Physics, Perth*, 2022.
4. **Sampson, M.** *et. al,*. “Simulating large scale cosmic ray propagation through compressible magnetohydrodynamic turbulence”. *Australian Institute of Physics Summer Conference, Brisbane*, 2021.
5. **Sampson, M.** *et. al,*. “Rapid and Effective Population Calibration with modified MCMC”. *Australian National University Student Conference, Canberra*, 2020 (best science talk prize).
6. **Sampson, M.** *et. al,*. “The impact of pair-instability mass loss on the binary black hole mass distribution”. *Australian National Institute for Theoretical Astrophysics Conference, Melbourne*, 2019.

## Conference Reviewing

1. NeurIPS *Workshop for the Physical Sciences*, 2023

## Honors and Awards

- 2022 Princeton Gradute Program First Year Fellowship in Natural Sciences
- 2021 Bok Honours Scholarship in Astrophysics at the RSAA
- 2020/2019/2018 Deans List of Commendation for academic excellence QUT
- 2019 ICRAR-Pawsey Centre Vacation Scholar
- 2018 Swinburne Centre for Astrophysics and Supercomputing Summer Vacation Scholarship

## References

Provided as per request